

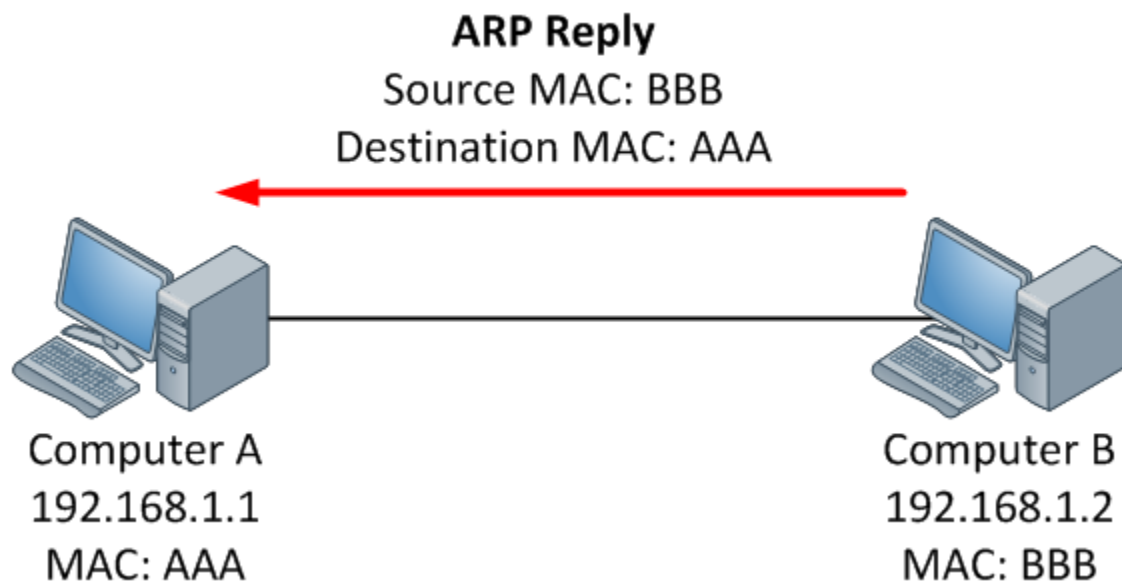
ARP and ARP POISONING



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What is a ARP?

ARP is a protocol used for mapping an IP address to a computer connected to a local network LAN .since each computer has a unique physical address called a MAC address, the ARP convert the IP address to the MAC address.



• TYPES OF ARP:-

- 1) Proxy ARP
- 2) Gratuitous ARP
- 3) Reverse ARP
- 4) Inverse ARP



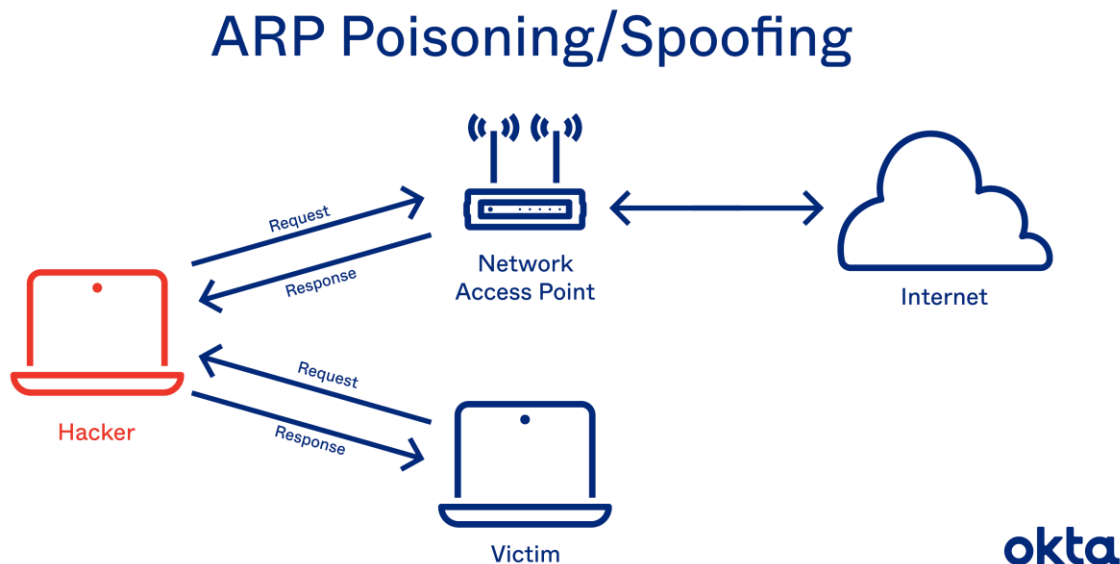
ARP Attacks:-

A malicious developer hoping to gain access to important data, could expose vulnerabilities and sneak inside, and you may never know it's happening.

Two types of ARP attacks exist.

ARP spoofing: A hacker sends fake ARP packets that link an attacker's MAC address with an IP of a computer already on the LAN.

ARP poisoning: After a successful ARP spoofing ,hacker changes the company's ARP table, so it contains falsified MAC MAPS.



- **How does address resolution Protocol (ARP) works:-**

Most of the computer programs/applications use **logical address (IP address)** to send/receive messages, however, the actual communication happens over the **physical address (MAC address)** i.e from layer 2 of the OSI model. So our mission is to get the destination MAC address which helps in communicating with other devices. This is where ARP

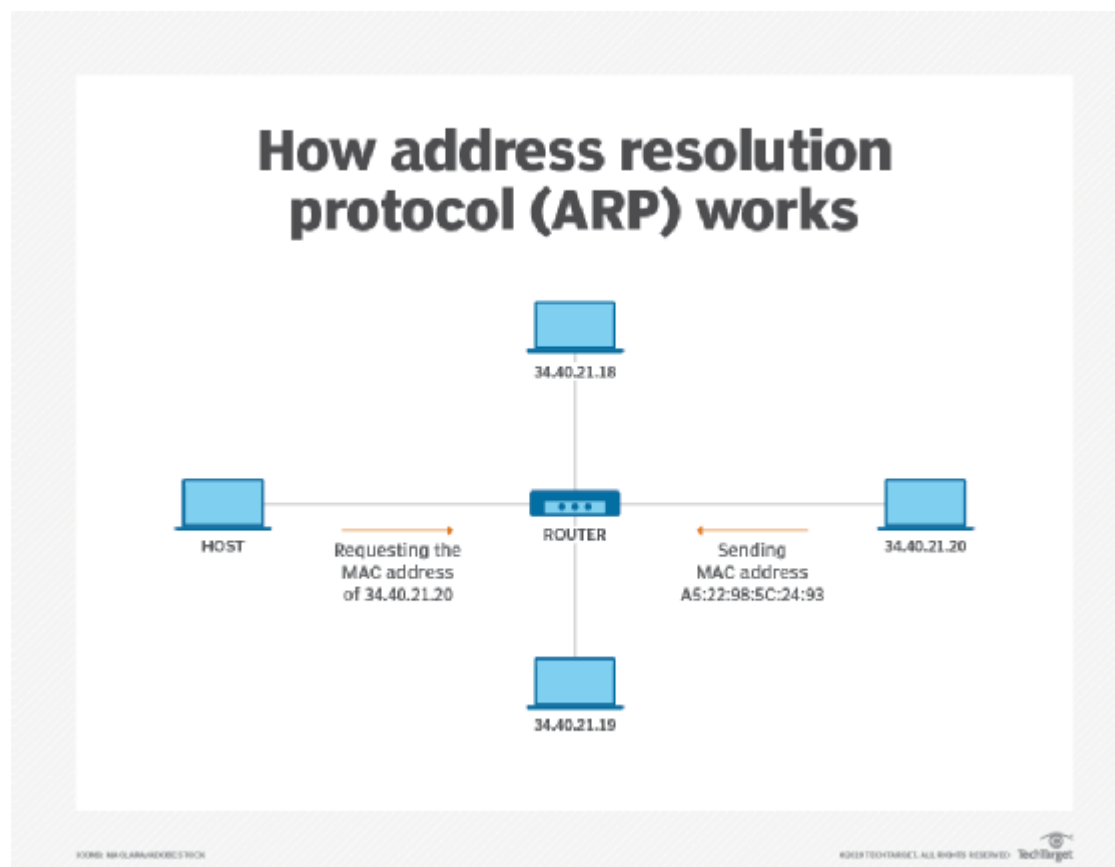
comes into the picture its functionality is to translate IP address to physical addresses. The acronym ARP stands for **Address Resolution Protocol** which is one of the most important protocols of the Network layer in the OSI model.

Note: ARP finds the hardware address, also known as Media Access Control (MAC) address, of a host from its known IP address.

Let's look at how ARP works.

Imagine a device that wants to communicate with the other over the internet. What ARP does? It broadcast a packet to all the devices of the source network. The devices of the network peel the header of the data link layer from the **protocol data unit (PDU)** called frame and transfer the packet to the network layer (layer 3 of OSI) where the network ID of the packet is validated with the destination IP's network ID of the packet and if it's equal then it responds to the source with the MAC address of the destination, else the packet reaches the gateway of the network and broadcasts packet to the devices it is connected with and validates their network ID.

1. The source device will broadcast the ARP request message to the local network.
2. The broadcast message is received by all the other devices in the LAN network. ...
3. The device whose IP address has matched with the destination IP address in the packet will reply and send the ARP Reply message.



- **What is ARP POISONING?**

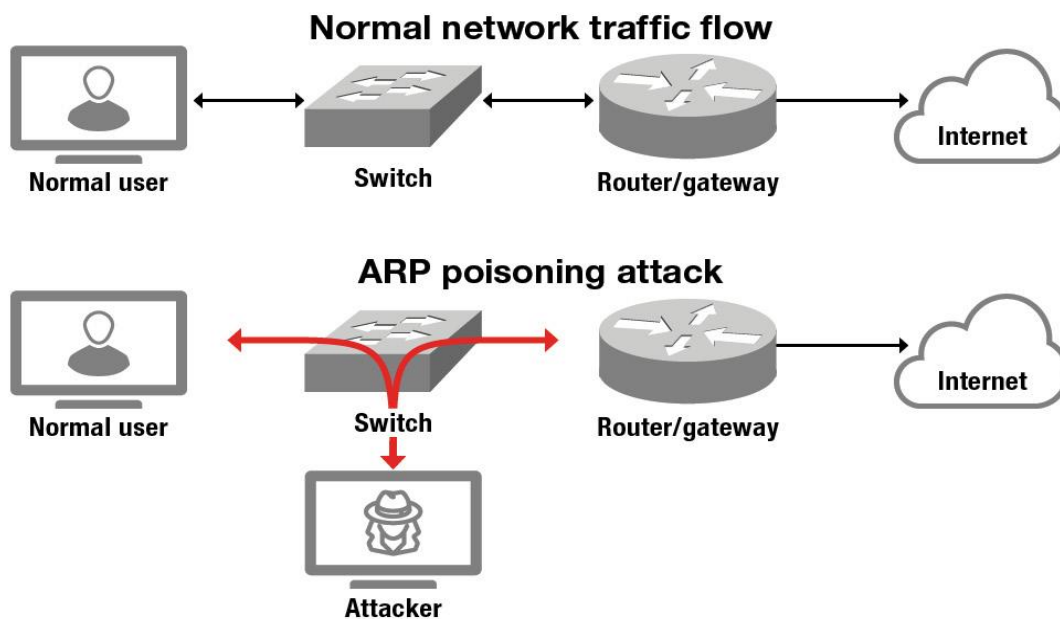
An ARP Poisoning is a Man in the middle attack that allows attackers to intercept communication between network devices.

- **The attack works as follows:**

1. The attacker must have access to the network. They scan the network to determine the IP addresses of at least two devices. Let's say these are a workstation and a router.
2. The attacker uses a spoofing tool, such as Arp spoof or Driftnet, to send out forged ARP Responses.

ARP Poisoning attack Example:-

If the table contains two different IP addresses that have the same MAC address, this indicates an ARP attack is taking place. Because the IP address 192.168. 5.1 can be recognized as the router, the attacker's IP is probably 192.168. 5.202.



Source: Crowe analysis

- **Types of ARP Poisoning Attacks**

There are two general ways in which an ARP Poisoning attack can occur: The attacker can either wait to see ARP requests for a particular target and issue a response, or send out an unsolicited broadcast message known as a "gratuitous ARP". The first approach

is less noticeable on the network, but potentially less far-reaching in its impacts. A gratuitous ARP can be more immediate and impact a greater number of victims but comes with the downside of generating lots of network traffic. In either approach, the corrupted ARP cache(s) on victim machines may be leveraged to further ends:

1) Man-in-the-Middle (MiTM) Attack

[MiTM attacks](#) are probably the most common, and potentially most dangerous, goal of ARP poisoning. The attacker sends out falsified ARP responses for a given IP Address, typically the default gateway for a particular subnet. This causes victim machines to populate their ARP cache with the MAC address of the attacker's machine, instead of the local router's MAC address. Victim machines will then incorrectly forward network traffic to the attacker. Tools like Ettercap allow the attacker to act as a proxy, viewing or modifying information before sending the traffic to its intended destination. To the victim, everything may appear normal.

Marrying ARP Poisoning with [DNS Poisoning](#) can dramatically increase the effectiveness of a MiTM attack. In this scenario, a victim user might type in a legitimate site such as google.com and be given the IP address of the attacker's machine, rather than the rightful address.

2) Denial of Service (DoS) Attack

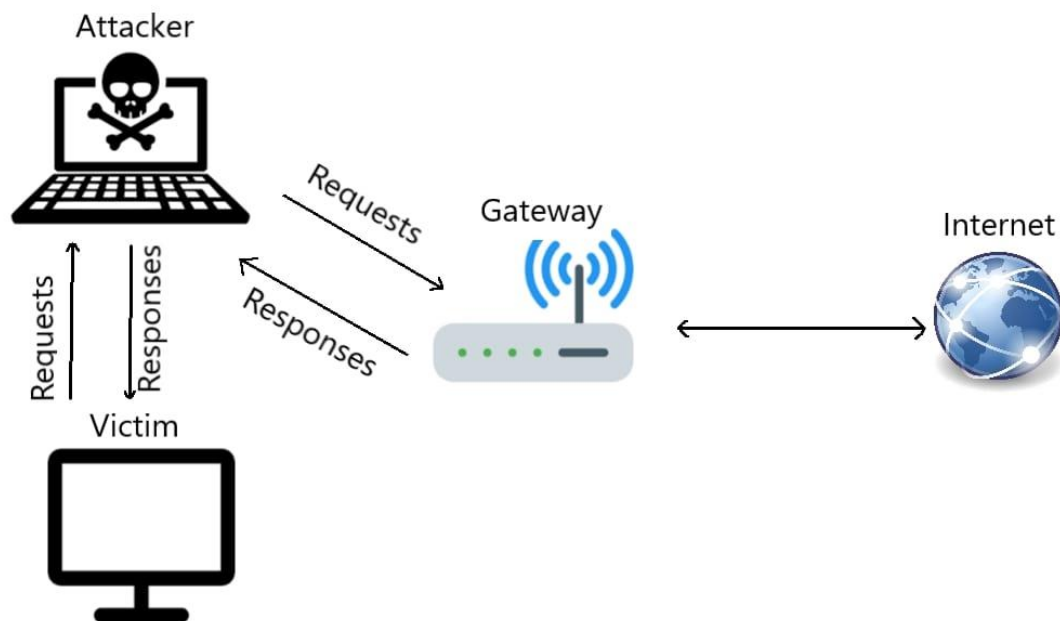
A DoS attack is aimed at denying one or more victims access to network resources. In the case of ARP, an attacker might send out ARP Response messages that falsely map hundreds or even thousands of IP addresses to a single MAC address, potentially overwhelming the target machine. This type of attack, sometimes known as ARP flooding, can also be used to target switches, potentially impacting the performance of the entire network.

3)Session Hijacking

Session Hijacking attacks are similar in nature to Man-in-the-Middle, except that the attacker will not directly forward traffic from the victim machine to its intended destination. Instead, the attacker will capture a genuine TCP sequence number or web cookie from the victim and use it to assume the victim's identity. This could be used, for instance, to access a target user's social media account if they happen to be logged in.

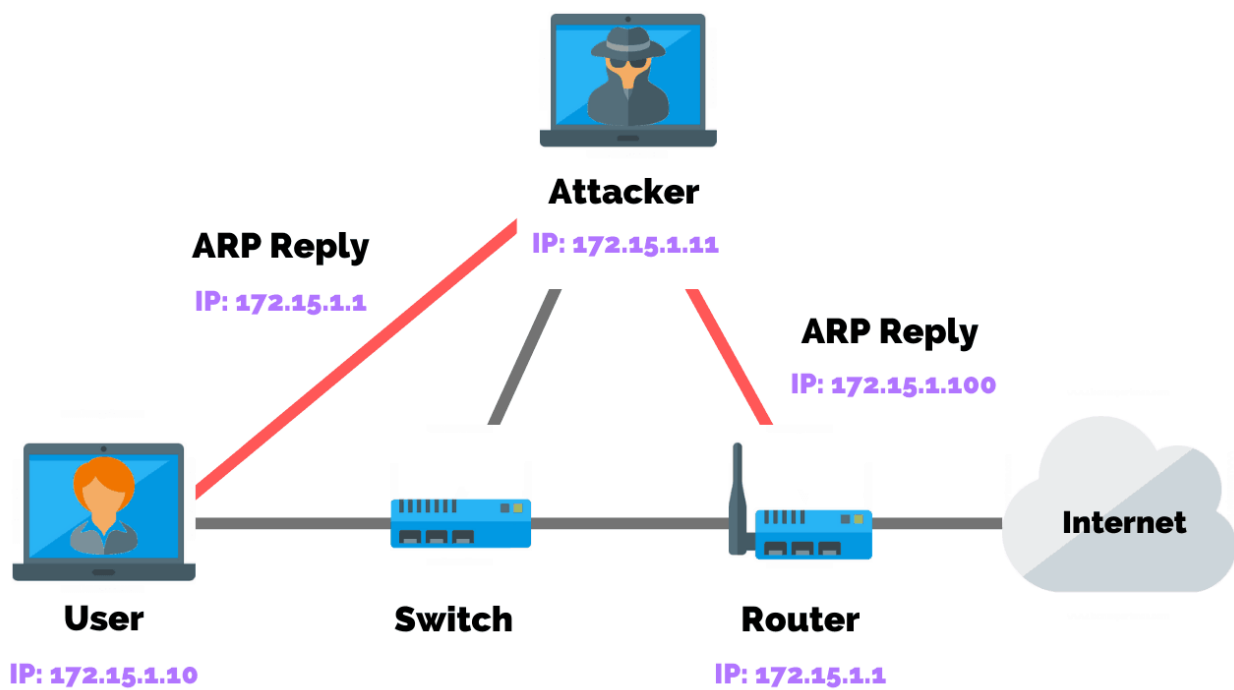
Working of ARP and ARP Poisoning

ARP Poisoning (also known as ARP Spoofing) is a type of cyber attack carried out over a Local Area Network (LAN) that involves sending malicious ARP packets to a default gateway on a LAN in order to change the pairings in its IP to MAC address table. ARP Protocol translates IP addresses into MAC address.



How to Detect an ARP Cache Poisoning Attack?

Hackers use a predictable series of steps to take over a LAN. They send a spoofed ARP packet, they send a request that connects to the spoof and they take over. The request is broadcast to all computers on the LAN, and control is complete.



- **Network administrators can use two techniques to detect ARP spoofing:-**
 1. **Passive:** Monitor ARP traffic and look for mapping inconsistencies.
 2. **Active:** Inject falsified ARP packets into the network. A spoofing attack like this helps you identify weak system.

